

KARAN CHAWLA

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EXPERIENCE

FIRST Robotics Competition

July 2021 – June 2025

Robotics Team Captain & Software Development Lead

Ottawa, ON

- Founding member and leader of cross-functional team, 90+ high school students to compete at a national level, qualifying for World Championships 2023 and Finalists at Provincial Championship 2023 & 2024
- Improved autonomous routine success accuracy to 95% by developing and programming teleoperated and autonomous routines in Java, leveraging simulations, tracked odometry, and computer vision.
- Achieved 50% faster movements and 70% greater positional accuracy by enhancing robot performance through the implementation of feedforward and PID feedback control systems in competition robots.
- Mentored younger students and fostered collaboration to build sustainability in the team.

Bruce Lab — University of Ottawa

July 2024 – Aug. 2024

Software + Controls Engineer

Ottawa, ON

- Reduced transient time by 80% and steady-state error by 40% by implementing PID control systems in C++ to stabilize current and voltage in platinum-nickel electrodeposition for Fuel Cell Electric Vehicles.
- Reduced calibration time by 300% by developing a Python user interface to conduct trials autonomously.
- Expanded trial voltage limits by 500% by refining an Arduino-based potentiostat electric circuit.

Canadian Robotics and Artificial Intelligence Ethics Design Lab

July 2024 – Aug. 2024

Robotics Research Lab Assistant

Ottawa, ON

- Accelerated project timeline by 50% by characterizing DJI's Robomaster S1 in a simulative environment.
- Achieved 1-second target tracking by implementing computer vision models with Python machine learning.
- Collaborated with supervisors and peers to adapt solutions in fast-paced, research environments.

PROJECTS

HackTheNorth WINNER — LipsLipsRevolution | Next.js, Tailwind, SymphonicLabs, MongoDB September 2024

- SymphonicLabs API integration to generate words from lip movement
- Hackathon-wide lip sync battle with updating frontend leaderboard, generating 50+ users
- Used advanced embedding model algorithm to determine similarity score
- Reduced latency from 30s to 7s by reverse-engineering and leveraging a faster WebSocket API.

Echo - An Agentic Voice AI Model for Getting Help — TerraHacks | React, Express.js, VAPI

August 2025

- Developed a full-stack web application using Express.js with React as the frontend
- Implemented VAPI and Twilio to call on the user's behalf
- Generates a transcription and recreated recording of the call for users

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks: Next.js, Flask, TailwindCSS, WPILib, Express.js, FastAPI, MongoDB, Git

Cloud Platforms: Google Cloud Platform, Firebase, Supabase

Libraries: React, pandas, NumPy, Matplotlib

AWARDS

HackTheNorth 2024 2x Award Winner: Best Use of MongoDB, SymphonicLabs

2x FRC Provincial Finalists: FIRST Robotics 2023, 2024

Harvard-MIT Math Tournament: Qualified Participant

FIRST Alumni Entrance Scholarship: Value of \$15,000

Dean's List Semi-finalist: First Robotics Competition

EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science — Engineering Science

Sep. 2025 – April 2030

International Baccalaureate Graduate

Ottawa, ON

Merivale High School — Grade: 97.7% + 41/45 IB Score

Sep. 2023 – June 2025